3.3.2 Number of research papers in the journals notified on UGC website during the last five years (10)

S.NO	ACADEMIC YEAR	No. of research papers in the journals notified on UGC website during the last five years
1	2020-2021	20
2	2019-2020	25
3	2018-2019	43
4	2017-2018	59
5	2016-2017	32
	Total	179

Principal

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Numerical Analysis of A Radiator Performance Using Aluminum Oxide Nanofluid

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Abstract- Most internal combustion engine to prevent overheating of the engine uses radiator as a heat exchanger to increase the heat transfer rate. The radiator are controls the engine temperature and thus increases engine efficiency .Liquid cooling requires the best exchange between the metal and the liquid for most efficient heat transfer. Cooling is one of the significant method of an automobile that affects the engine performance. The present study is to analyze numerically, the forced convection heat transfer of Aluminum oxide (Al_2O_3) nanoparticles dispersed in water as a base fluid circulating in the radiator. Three different concentrations of nanofluids 0.1%, 0.5% and 1% have beenused.The flow rate of water based nanofluid through the pipe is 0.138 kg/s to 0.555kg/s.The free stream temperature of air is varied from 20°C to 40°C.The simulation results shows that the presentation of Al_2O_3nanofluid with low concentrations can increase heat transferrate compared with purewater,

Keywords:Aluminum oxide nanofluid, temperature drop, Car radiator, Heat Transfer, Ansys.

1. INTRODUCTION

The radiators are used in water cooled automobile engines to keep water rotation in the Cylinder heads of engines at comparatively constant temperature and thus avoidhigh temperature of the engine. Liquid cooling involves the best conversation between the metal and the liquid for best efficient heat transfer. The coolant is generally water-based,

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International Journal & Magazine of Engineering, Technology, Management and Research

A Peer Reviewed Open Access International Journal

Performance Analysis of an Energy Efficient Level Shifter Using MTCMOS Technology

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ABSTRACT

The level shifters play a crucial role in the circuits and systems with multi supply voltages. In this paper, level shifter is proposed which is energy efficient, in order to achieve the above threshold voltage from the sub-threshold voltage conversion. It is a circuit with hybrid structure consisting of the cross- coupled level shifter and Wilson current mirror circuit. Significantly, the leakage power is reduced by addressing the voltage drop issue of the level shifter based on the Wilson current mirror, with the advantage of wide input voltage range for the Wilson current mirror level shifter. In addition to that, the multi-threshold CMOS (MTCMOS) technology is introduced to provide flexibility for our ultra-low power design. The simulation results were obtained by simulating the design using 65nm CMOS process which validates our proposed implementation and without the need of any intermediate power supply, an ultra-low power consumption of 5.6 uJ per conversion from 0.2 V to 1 V at 1MHz is achieved.

I. INTRODUCTION

Multi-supply voltage domain technique is effective method to reduce power dissipation. In this technique, the design is partitioned into separate voltage domains, where each domain is operating at distinct power supply levels. Voltage level shifter is a circuit which converts the signal from one voltage level to another. To interface various domains in multi-supply voltage design, voltage level shifter is used. At the boundaries of different voltage islands on the system-on-chip (SOC) voltage level shifter is used. Voltage level shifter is a device which converts one voltage level to another. Voltage level shifters are used to interface various circuit blocks operating at different supply voltages. The multi-threshold voltage CMOS technique is used in the design of voltage level shifter in order to reduce delay and power. Low-threshold voltage devices are used to reduce delay and power dissipation is reduced with the use of high threshold voltage devices.

II. CONVENTIONAL METHODS

At present there are two conventional designs that are proposed for the implementation of level shifter: the cross coupled structure and the current mirror structure.

II. I CROSS-COUPLED STRUCTURE:

The cross coupled level shifter in Fig. 1 (a) takes significant advantage of the full output swing generated by a positive feedback. The major limiting issue for this design is that the driving strength for the NMOS pair (MN1 and MN2) is much weaker than that of the PMOS pair, when the input is below the threshold voltage of the NMOS pair. This result in the failure of the output's toggling. Although the driving strength can be improved by greatly upsizing the NMOS pair but results in large area and load capacitance.

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IJESRT
INTERNATIONAL JOURNAL OF ENGINEERING SCIENCES & RESEARCH TECHNOLOGY

DESIGN OF CONTROLLER FOR POWER QUALITY IMPROVEMENT BY UPQC

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ABSTRACT

Active Power Filters can be classified, based on converter type, topology and the number of phases. An artificial neural network (ANN), often just called a "neural network" (NN), is a mathematical model or computational model based on biological neural networks. It consists of an interconnected group of artificial neurons and processes information using a connectionist approach to computation. In most cases an ANN is an adaptive system that changes its structure based on external or internal information that flows through the network during the learning phase.

In this paper, various extraction algorithms for generating reference signals and various modulation techniques for generating pulses already developed and published are discussed. Criterion for selection of dc link capacitor and interfacing filter design are also discussed. The Objective of this paper, one such APLC known as Unified Power Quality Conditioner (UPQC), which can be used at the PCC for improving power quality, is designed, simulated using proposed control strategy and the performance is evaluated for various nonlinear loads (steel plant loads)

KEYWORDS: Power Quality, UPQC, ANN, PCC.

1. INTRODUCTION

Power electronic based power processing offers higher efficiency, compact size and better controllability. But on the flip side, due to switching actions, these systems behave as non-linear loads [1-3]. Therefore, whenever, these systems are connected to the utility, they draw non-sinusoidal and/or lagging current from the source. As a result these systems pose themselves as loads having poor displacement as well as distortion factors. Hence they draw considerable reactive volt-amperes from the utility and inject harmonics in the power networks. Loads, such as, diode bridge rectifier or a thyristorbridge feeding a highly inductive load, presenting themselves as current source at point of common coupling (PCC), can be effectively compensated by connecting an APF in shunt with the load [4-6]. On the other hand, there are loads, such as Diode Bridge having a high dc link capacitive filter. These types of loads are gaining more and more importance mainly in forms of AC to DC power supplies and front end AC to DC converters for AC motor drives. For these types of loads APF has to be connected in series with the load [4, 7].

The voltage injected in series with the load by series APF is made to follow a control law such that the sum of this injected voltage and the input voltage is sinusoidal. Thus, if utility voltages are non-sinusoidal or unbalanced, due to the presence of other clients on the same grid, proper selection of magnitude and phase for the injected voltages will make the voltages at load end to be balanced and sinusoidal.

The shunt APF acts as a current source and inject a compensating harmonic current in order to have sinusoidal, in-phase input current and the series APF acts as a voltage source and inject a compensating voltage in order to have sinusoidal load voltage. The developments in the digital electronics, communications and in process control system have increased the number of sensitive loads that require ideal sinusoidal supply voltage for their proper operation. In order to meet limits proposed by standards it is necessary to include some sort of compensation. In the last few years, solutions based on combination of series active and shunt active filter have appeared [8-9]. Its main purpose is to compensate for supply voltage and load current imperfections, such as

sags, swells, interruptions, imbalance, flicker, voltage imbalance, harmonics, reactive currents, and current unbalance [10-16]. This combination of series and shunt APF is called as Unified Power Quality Conditioner (UPQC). In most of the articles control techniques suggested are complex requiring different kinds of transformations. The control technique presented here is very simple and does not require any transformation.

3.3.2-95

Realisation of QCA Based Adder Circuit for High Speed Application

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Abstract- In the existing technologies, Quantum-dot Cellular Automata (QCA) is recently proposed alternative method in the CMOS design. It is one of the best method to design high speed digital and ultralow-power circuits. Several binary and arithmetic circuits are implementing using QCA technologies, but still some improvements are possible if the logic gates are constitutionally accessible in these technologies.

This short proposes a new procedure to design QCA-based BCD adders. In this paper ripple carry adder using majority gates as a existed system and brunt kung adder using majority gate as a proposed system. Utilizing new logic assembling and purpose-designed QCA modules, area and consumption of power significantly lower than existing system.

Key Words—Quantum-dot Cellular Automata (QCA), BCD adders, decimal arithmetic.

1. INTRODUCTION

QCA technology has become one of the most attractive approaches for the development of next-generation ultra dense low-power high-performance digital circuits. Among the logic circuits recently proposed, arithmetic sub modules represent examples of the most investigated structures. Independently of the performed logic function, the modules are designed smartly combining inverters and majority gates (MGs), these are the only fundamental logic gates available within the QCA technology. Decimal arithmetic has recently a great deal of attention since several financial, commercial, and Internet based applications increasingly require higher precision.

This brief proposes a new approach to design QCA-based technologies n-digit (with $n \geq 1$) BCD parallel adders are able to achieve increasing speed higher than present counterparts without sacrificing

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3.3.2 - 97

A Novel Approach to Implement MDC FFT Architecture For Low Power Applications

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Abstract- The 4G AND 5G wireless communications is dominated by MIMO-OFDM which defines as multi input and multi output orthogonal frequency division multiplexing. The multiple signals over multi antennas can be send or capable through this multi input and multi output and the radio channels can be divided into largely spaced sub channels through orthogonal frequency division multiplexing. The communication system uses this data without loss in reliability. In previous time division multiple access, code division multiple access are used with a combination of MIMO. But in spite of all these MIMO with OFDM is much famous. It is because of its high data rate, high message deliver capacity, high throughput. .At wires LAN and some standard networks at mobile communications it is famous for these reasons. If the data size increases then the memory size also increases rapidly in MDC based MIMO- OFDM, but in simplest manner data flow can be controlled by using multipath delay commutator. Simplicity is the main reason for implementing MIMO based OFDM and in order to increase reliability the bigger obstacles can be eliminated, it is possible by making the user data into a closely spaced narrow sub channels.

To implement fast Fourier transform (FFT) processors for multiple input multiple output-orthogonal frequency division multiplexing (MIMO-OFDM) systems with variable length, this Project presents a multipath delay commutator (MDC)-based architecture and memory scheduling. The fft/ifft processor can be implemented based MIMO-OFDM system based on the MDC architecture. RAM, FIFO, input buffer and output sorting buffer are implemented using this design and by using XILINX ISE 12.3i the functionality verification and the synthesis are carried out and the reduced delay values are showed.

Index Terms— Orthogonal frequency division multiplexing (OFDM), memory scheduling, fast Fourier transform (FFT), multiple-input and multiple-output (MIMO), pipeline based multipath delay commutator (MDC), WiMAX, output sorting,.

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3.3.2 - 99

Efficient Mirror Adder Design using Quantum Dot Cellular Automata

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Abstract- Quantum dot cellular automata (QCA) are a growing nano technology for the development of high performance ultra-dense low power digital circuits. QCA based numerous effective binary and decimal arithmetic circuits are implemented, however important improvements are still possible. These works determines Mirror Adder circuit design with CMOS and compare it with QCA mirror adder design. We present proportional study of mirror adder cells designed using conventional CMOS technique and mirror adder cells designed using quantum-dot cellular automata. QCA based mirror adders are superior in terms of area

I. INTRODUCTION

Quantum Dot cellular Automata (QCA) do not use transistors. QCA design addresses the concern of device density and interconnection. The elementary element of QCA is a quantum cell. Each quantum cell consumes electrons in them wherever electron diffusion take place on the columbic interaction of the electrons. QCA is an advanced research program and exertions are made to reduce the complexity of the circuits. When included size is compact to nanometers than Quantum effects such as Tunneling takes place [1].QCA circuits can be straight gained from conventional designs with addition of special clocking system. This offers very easy diffusion of predictable circuits to get transformed into QCA structures. QCA structures are designed as an array of quantum cells, where every cell has electrons in them where electrostatic interaction with its adjacent cells takes place. QCA uses a new technique for reckoning. It uses polarization upshot rather than predictable current for the transmission of information which contains the digital information. Thus a cell is accountable for the transfer of information all the way over the circuit. The basic operators used in QCA are three input majority gates and invertor. This study offers the design of different types of adders. Hence the planned design is used to reduce the area and complexity. Current transistor-based semiconductor devices are becoming resilient to scaling. For transistor circuit, the power rakishness due to leakage current is a big problem [4-7]. A possible substitute to these problems is Nanotechnology based Quantum-dot cellular automata (QCA) circuits [7-8]. The emphasis of this paper is on project of efficient Mirror adder circuit [3]. Prior to our work on mirror adder we scrutinized some prior works on adder circuits in QCA technology. J. Cocorullo, Giuseppe planned Design of Efficient BCD Adders in Quantum Dot Cellular Automata [4]. An Efficient Design of Full Adder in Quantum-Dot Cellular Automata (QCA) Technology is shown by M. Mohammadi, S. Gorgin [9]. Efficient Design of a Hybrid Adder in Quantum-Dot Cellular Automata is presented by V. Pudi and K. Sridharan [12]. M. Gladshtein

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The Efficient Enhanced Methodology for Multidimensional Packet Classification

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Abstract-Wireless Sensor networks of sensor nodes are imagined to be sent in the physical condition to monitor a wide assortment of true marvels. A wireless sensor network can get isolated into various associated parts because of the disappointment of a separation of its nodes, which is known as a "cut." Here, in this paper, issue of identifying cuts by the rest of the nodes of a wireless sensor network has been referenced. Additionally pursued DCD algorithm that enables each node to distinguish when the network to a uniquely assigned node has been lost, and at least one nodes to recognize the event of the cut. This algorithm is appropriated and non-concurrent. The Distributed Cut Detection (DCD) algorithm proposed here likewise empowers a subset of nodes that encounter CCOS occasions to distinguish them and locate the inexact area of the cut as a rundown of dynamic nodes that lie at the boundary of the cut. The upside of DCD algorithm is that assembly rate of the iterative plan is very quick and autonomous of the size and structure of network

Keywords: Boundary Cutting, Adaptive Binary Cutting, access control, firewalls, intrusion detection, and policy based routing.

I. INTRODUCTION

The way toward arranging bundles into "streams" in an Internet switch is called parcel characterization. All bundles having a place with a similar stream comply with a predefined rule and are prepared along these lines by the switch. Bundle order is an empowering capacity for an assortment of web applications including Quality of administration (QoS), security, observing, mixed media Communications [1]. Developing and changing network activity prerequisites conjures need of bigger channel with increasingly complex guidelines, which thusly offers ascend to various quick bundle characterization algorithms. Parcel grouping is required for non-best-exertion administrations, for example, firewalls and interruption recognition, switches, ISPs and normally in the most calculation escalated undertaking among others. Administrations, for example, data transfer capacity the board, movement provisioning, and usage profiling likewise rely on bundle characterization. Bundle comprises of header and data information and header comprises of MAC address, IP address, port number and so on. Generally, the Internet gave just a "best-exertion" benefit, treating all parcels heading off to a similar goal indistinguishably, adjusting them in a first start things out served way. Be that as it may, web clients and their requests for various quality administrations are expanding step by step. Along these lines, Internet Service Providers are looking for approaches to give separated administrations (on a similar network framework) to various clients dependent on their distinctive necessities and desires for quality from the Internet. For this, switches need the capacity to recognize and disconnect movement having a place with various streams. The capacity to arrange every approaching bundle to decide the stream it has a place with is called parcel characterization and could be founded on a subjective number of fields in the parcel header. Parcel order is a multi-dimensional type of IP query and finding longest prefix coordinating to give next-jump in routers.[2] Number of bundle grouping algorithms have been proposed starting late, most of them remain in logical examination and/programming reenactment stage, and few of them have been realized in business things as a non-explicit game plan. The hole among hypothesis and practice in existing work can be compacted by various

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Machine Learning Applications for Sentiment Analysis Aspect based Opinion Mining

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Abstract- Presently multi day's online consumer review is most incredible asset for decision making. This term fill in as electronic verbal (EWM) which turn into an inexorably famous. A large number of individuals are currently purchase items and services by means of online. Web services are given element to clients straightforwardly. The web can gives a broad wellspring of consumer reviews. The client can peruse every one of the reviews and assess reasonable view about item or administration. This can material just for predetermined number of reviews displayed on web. The web contain more than many reviews then issue arrived and tedious too. A content preparing system is alluring which abridge every one of the reviews. This system would be discover general viewpoint classification tended to in all review sentences. The technique introduced in this system which applies affiliation rule mining on co-event recurrence information to discover these angle classifications. From this outcome, produce extremity score for every perspective class. This extremity score assesses reasonable decision making for customer and additionally organization.

Keywords— sentiment analysis, polarity, aspect, opinion mining, Feature extraction, machine learning technique,

I. INTRODUCTION

Information mining is the way toward finding designs in colossal informational collections including strategies at the crossing point of machine learning, insights, and database frameworks. Information mining is an interdisciplinary subfield of software engineering with a general question mine data (with insightful strategies) from an informational index and change the data into an understandable structure for further use. Information mining is the investigation venture of the "learning revelation in databases" process, or KDD. Beside the crude analysis step, it likewise includes database and information the executive's viewpoint, information preprocessing, model and induction generosity, intriguing quality measurements, intricacy thought, post-preparing of found structures, representation, and online refreshing. The expression "information mining" is in reality a misnomer, in light of the fact that the point is the extraction of examples and learning from a lot of information, not simply the extraction (mining) of information. It likewise is a popular expression and is regularly connected to any type of huge scale information or data preparing (gathering, extraction, warehousing, analysis, and insights) and additionally any case of PC decision emotionally supportive network, and also man-made consciousness (e.g., machine learning) and business knowledge. The genuine information mining errand is the self-loader or routine analysis of huge amounts

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Annoyed Methodology Designed For Diminishing Data Transmission in Ip Networks

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Abstract- A probabilistically correlative failure (PCF) model is developed to quantify the impact of IP link failure on the reliableness of backup methods. Vitality utilization of system hardware and environment security are increasing expanding concern as of late, to create vitality proficient system structural engineering and operational expenses as to decrease the vitality utilization of the web. In cross layer methodology go down ways are chosen and generally utilized as a part of IP systems to shield IP joins from disappointments. Subsequently cross layer methodology for minimizing directing disturbance brought about by IP join disappointments and advancing the vitality amid this operation will be tended to. In the present examination, we focus on minimizing the vitality utilization of an IP over WDM systems and minimizing steering interruption created by IP join disappointments amid cross layer methodology, with the CPF model we create effective calculation to choose different solid reinforcement ways to ensure each IP join, when an IP connection comes up short, its movement is part onto numerous reinforcement ways and the rerouted activity load on go down ways and the rerouted activity load on each IP connection does not surpass the usable transmission capacity, we unravel our methodology utilizing genuine ISP system with both optical and IP layer topologies particularly in IP over WDM System vitality is devoured by system components at both IP and WDM layers. What's more, current system base have no Vitality sparing plan, here we create effective methodology called blended whole number direct programming(MILP).this methodology is taking into account customary virtual topology and activity preparing outlines, trial results demonstrate that two reinforcement ways are sufficient for securing a coherent connection, and move down way chose by our methodology are no less than 18% more dependable and the steering interruption is lessened by no less than 22 percent, and the proposed methodology keeps the rerouted movement from meddling with typical movement, at long last it is

Additionally valuable and intriguing to discover a vitality productive system configuration is likewise an expense proficient configuration due to the way that IP switches assume a critical part in both vitality utilization and system cost in the IP over WDM systems.

Keywords - Routing, failures, cross-layer, recovery. Network security

I. INTRODUCTION

IP join disappointments are genuinely regular in the Internet for different reasons. In rapid IP systems like the Internet spine, detachment of a connection for a few seconds can prompt a large number of bundles being dropped. In this manner, rapidly recouping from IP join disappointments is critical for improving Internet unwavering quality

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Data Sharing towards Integrated Secure Key Encryption System with Big data in Cloud Computing

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Abstract- These day's Cloud computing developed as a practical and demonstrated conveyance stage for giving business or purchaser IT services over the Internet. Be that as it may, the services given by the cloud are of outsiders, experiencing security and protection of the clients BIG Data is basic at cloud storage. This paper displayed a methodical report on the feasibility of elliptic bend cryptography in anchoring BIG Data at cloud storage. Based on the deliberate investigation a protected gathering correspondence structure is additionally displayed in this work. This work accept the cloud storage is raised with Hadoop based data focus. So as to do efficient investigation private key calculations (EC, RSA), open key calculations (AES, DES) and half and half calculation (Elliptic bend Dephihelmen ECDH) are considered to check the BIG Data security at cloud based data focus. From the trial results this examination prescribes that for the unbound key sharing channel Elliptic Curve (EC) based cryptographic calculations are very secure than RSA and private key calculations.
Keywords: Cloud Computing, Cryptography, Data Sharing, Encryption System, ABE, Cloud security, Encryption and Decryption Algorithm.

I. Introduction

Big data is a term that depicts the mind boggling or large volume of organized and unstructured data. It is hard to process utilizing available database the executive's instruments. To separate vital quality from Big Data, we require perfect preparing strategies, explanatory capacities and aptitudes. There are a few big data preparing methods including cloud condition accessible. In the latest two decades, the reliable augmentation of computational power has made an amazing stream of data. Huge data is ending up being progressively available and also increasingly reasonable to PCs. For example, the celebrated relational association, Facebook, serves 570 billion webpage visits for consistently, stores 3 billion new photos every month, and supervises 25 billion bits of substance. Google's request and notice business, Facebook, Flickr, YouTube, and Linkedin use a store of synthetic mental aptitude traps; require parsing boundless measures of data and settling on decisions quickly. Sight and sound data mining stages make it basic for everybody to achieve these destinations with the base proportion of effort to the extent programming, CPU and framework. All of these delineations exhibited that staggering huge data challenges and basic resources were administered to reinforce these data genuine activities which provoke high accumulating and data taking care of costs. The present advances, for instance, matrix and cloud computing have all normal to get to a ton of totaling in order to enroll compel resources and offering a lone structure see. Among these advances, cloud computing is transforming into a skilled development modeling to perform considerable scale what's increasingly, complex computing, and has vexed the manner in which that enlisting establishment is distracted and used. Likewise, a fundamental purpose of these advances is to pass on preparing as a response for taking care of gigantic data, for instance, large scale, multi-media and high dimensional data sets. Big data and cloud computing are both the speediest moving progressions recognized in Gartner Inc's. 2012 Hype Cycle for Emerging Technologies. Cloud computing [1] is

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Network Load Balancing Routing for Traffic Congestion Control Nonlinear Systems in Networks

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Abstract- This paper presents new system for load balancing in a network. Load balancing is a networking technique for disseminating workload over different figuring assets, for example, a cluster network joins, focal preparing units or plate drives. Load Balancing is generally given by committed programming or equipment, for example, a multilayer switch or Domain Name System Sever Process. There are different calculations to perform load balancing. In this paper we will talk about how to perform load balancing utilizing stores and examine the favorable position and impediments of utilizing this strategy to perform load balancing. The present time has seen enormous development of the Internet and different applications that are bolstered by it. There is a tremendous weight on Internet Service Providers (ISPs) to make accessible sufficient administrations for the deals like VoIP and Video on interest. Since the assets like figuring power, data transfer capacity and so forth are constrained, the traffic should be built to appropriately abuse them. Because of these restrictions, terms like Traffic Engineering, Quality of Service (QoS) appeared. Traffic Engineering comprehensively incorporates systems like multipath steering and traffic part to adjust the load among various ways. In this report, we review different methods proposed for load balancing that are accessible on the Internet. We here make an effort not to be thorough but rather break down the vital strategies in the writing. Present overview would provide another guidance to the exploration in this domain.
Keywords- Load Balancing; Heaps; Network, Traffic Engineering; Load Balancing; MPLS Networks.

I. INTRODUCTION

Load awkwardness is a guileful factor that can decrease the performance of a parallel application altogether. For a few applications, load is anything but difficult to anticipate and does not shift progressively. In any case, for a noteworthy class of utilizations, load portrayals by bits of calculations fluctuate after some time, and might be harder to foresee. This is getting to be expanding predominant with development of new-advanced applications. This new technique for load balancing utilizing piles will consequently and powerfully disperse traffic over numerous servers. It will empower us to accomplish more prominent dimensions of adaptation to internal failure and enhanced performance consistently giving the required measure of load balancing limit expected to circulate network traffic. Traffic designing (TE) comprehensively characterizes the improvement of useful capacities of the network [1]. This improvement is finished by occupying the traffic to the ways that are softly loaded so as to adjust the load among the ways according to the different measurements determined. Procedures for TE proposed everywhere throughout the world can be partitioned in to state dependant and time dependant. Time dependant functionalities build the traffic based on prolonged stretch of time scale. Then again, state dependant techniques adjust the traffic in brief time scale contingent upon the diverse measurements determined on the web or disconnected of the present traffic. The point of both these techniques for course is to adjust the traffic in order to stay away from the clog. Present day IP network depends on the best exertion benefit however as there is significant development of the applications that depend on the

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Cryptographic Data Encryption and Decryption for Secure data Transfer towards Network Security

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Abstract: Network Security and Cryptography is an idea to ensure Network and information transmission over remote Network. Information Security is the fundamental part of secure information transmission over inconsistent Network. Network security includes the approval of access to information in a Network, which is controlled by the Network overseer. Clients pick or are allocated an ID and secret word or other confirming data that permits them access to data and projects inside their power. Network security covers an assortment of PC Networks, both open and private, that are utilized in regular employments leading exchanges and interchanges among organizations, government offices and people. Networks can be private, for example, inside an organization, and others which may be available to free. Network security is associated with associations, endeavors, and different sorts of establishments. In this paper we additionally contemplated cryptography alongside its standards. Cryptographic frameworks with figures are portrayed. The cryptographic models and calculations are laid out.

Keywords: Network Security, Cryptography, Security Challenges.

I. Introduction

Network Security is the most indispensable part in data security since it is in charge of anchoring all data went through arranged PCs. Network Security alludes to all equipment and programming capacities, attributes, highlights, operational methodology, responsibility, measures, get to control, and regulatory and the board arrangement required to give a worthy dimension of insurance for Hardware and Software , and data in a network.Network security issues can be partitioned generally into four intently entwined regions: mystery, validation, nonrepudiation, and trustworthiness control. Mystery, additionally called classification, has to do with keeping data out of the hands of unapproved clients. This is the thing that typically rings a bell when individuals consider arrange security. Confirmation manages deciding whom you are conversing with before uncovering touchy data or going into a business bargain. Nonrepudiation manages marks. Message Integrity: Even if the sender and beneficiary can verify one another, they additionally need to protect that the substance of their correspondence isn't changed, either malicously or unintentionally, in transmission. Expansions to the checksumming procedures that we experienced in dependable transport and information interface conventions. Cryptography is a developing innovation, which is vital for Network security. The far reaching utilization of modernized information stockpiling, preparing and transmission makes touchy, important and individual data powerless against unapproved get to while away or transmission. Because of proceeding with headways in correspondences and listening in advances, business associations and private people are starting to secure their data in PC frameworks and Networks utilizing cryptographic strategies, which , until as of late, were solely utilized by the military and discretionary networks. Cryptography is a fundamental of the present PC and correspondences Networks, shielding everything from business email to bank exchanges and web

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Big Data Analytics Techniques and Approaches to detect Cyber Crime

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Abstract- This paper is about big data systems used to battle crime in cyber speculations. The fundamental difficulties in crime examination have been tended to and the kind of crimesters included have been contemplated in the light of various circumstances. The two principle kinds of data mining strategies i.e. administered and unsupervised learning techniques utilized in crime investigation have been seriously inspected with the assistance of existing writing. This is expected to help in perceiving which data mining systems help the most in crime examination in particular spaces. Banks are an indispensable piece of a nation's economy, adding to the two individuals and governments. In later past a great deal of crime exercises have been accounted for in banks due to individuals with personal stakes. This paper tosses light on normal insider crimes happening in banks and furthermore attempts to order them into various sorts. Here we give a point of view of definition, factors identified with such classes of crime and furthermore the difficulties one faces in identifying crimes. It is critical to identify such crime exercises consequently before it is past the point of no return and bring the general population or gathering of individuals into terms. Data mining system comes convenient as it recognizes irregular patters in given data set. This paper focusses on various nonexclusive data mining systems and in explicit, the procedures utilized for identifying insider crimes. We additionally drill down best accessible systems in acknowledging insider crimes with applicable outlines. As it is extremely apparent in this day and age, Big Data challenges are springing up all day every day, we have felt free to exhibited as a future improvement a couple of enormous data challenges while taking care of banking data.

Keywords – Data mining, fraud pattern detection, clustering, supervised learning, Cyber crime

I. Introduction

Cybercrime is any sort of wrongdoing that should be possible in, with, or against systems and PC frameworks [10] [3]. Cybercrime is getting expanded with the expanding dangers because of online crimes and exploitative hacking. With both data and digital security risk expanding, associations must be prepared to furnish themselves with foreseeing and forestalling cybercrime. Cybercrime specialists are utilizing huge data apparatuses to recognize the potential dangers and identify cybercrime occurrences like charge card crimes. Huge data investigation is empowering organizations to examine voluminous measure of data they assemble amid money related exchanges, any district explicit data and others too. Battling digital wrongdoing is of most extreme significance today because of expanded danger of digital robbery. Huge data instruments are being utilized to battle digital assaults. Huge Data examination can help distinguish crime and recognize burglary and can encourage advanced legal investigation. In this paper, a short overview is made about different systems utilized in examining huge data to recognize the crimes identified with charge cards by breaking down extensive arrangement of data. One point of this examination is to distinguish the client show that best recognizes crime cases. Along these lines, data mining is an essential system for every single movement of the charge card process. For instance, it tends to be utilized for ordering great clients or terrible clients which is completely founded on their application data and, likewise, distinguishing an abuse of a charge card dependent on buy data of a client. The effectiveness of anticipating the integrity or disagreeableness of a candidate can diminish credit danger of a Mastercard suppliers. While, if the supplier settle on any wrong choice by giving Visas to terrible clients, it will result in huge

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Decentralized E-bidding Governance Application using Blockchain

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Abstract- Due to technological advancements, traditional paradigms like central databases were challenged. Since society values like mutual understanding and honesty shifted, which implies the expectations of the government also transfers from traditional models to something new technologies which overcome those challenges. Some technologies such as internet play a crucial role on data and the activities between the users and government.

To enable integrity, non-repudiation, evidential along with immutability to the data requires the desirable technology to support the above the requirements. BLOCKCHAIN is a technology which is mainly used to facilitate crypto currencies as a record of transactions. A known example is Bitcoin, recent years BlockChain utility is being recognized through smart contracts. In this paper we propose a concept of smart contract on government tendering scheme. This scheme is implementing in Ethereum platform.

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Drone Assisted Artificial Rains

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Abstract- India is an agro-based country. It is estimated that seventy two percent of Indian economy is through agriculture. Within the past thirty years our country faced a lot of water scarcity not only for agriculture but also domestic, Industrial and commercial

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SYNTHESIS AND CHARACTERISATION COBALT FERRITE NANO COMPOSITE

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Abstract:
Polymer hybrids have become a major area of research and development owing to the remarkable properties and multifunctional behaviour deriving from their nanocomposite/nanohybrid structure. In this class, magnetic polymer nanocomposite are of special interest because of the combination of excellent magnetic properties, high specific area, surface active sites, high chemical stability and good biocompatibility.Crystalline, magnetic, cobalt ferrite nanoparticles were synthesized from an aqueous solution containing ammonium hydroxide as a capping agent by a thermal treatment followed by calcination at varioustemperatures. The structural characteristics of the calcined samples were determined by X-ray diffraction (XRD), Fourier transform infrared spectroscopy (FT-IR), and transmission electron microscopy (TEM). Magnetization measurements were obtained at room temperature by using a vibrating sample magnetometer (VSM), which showed that the calcined samples exhibited typical magnetic behaviors.
Key Words: Polymer Nano Composite, Alginate Cobalt Ferrite, Nano Composite,BET, TEM SEM, Core – Shell

Introduction:
Transition Metal Oxides

Transition metal oxides are technologically important materials that have found to be relevant in chemical applications.

Classifying metal oxides as catalysts are quite tedious since it involves a variety of crystal systems of different compositions with a wide range of physico-chemical properties. Oxide catalysts fall into two general categories. They are either poor electrical conductors or good conductors. Insulator oxides are those in which the cationic material has a single oxidation

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Development And Validation of An Rp-Hplc Method For The Determination of Balofloxacin In Human Plasma

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Abstract: A new simple, sensitive and stability indicating-HPLC method for the determination of Balofloxacin in pure and pharmaceutical dosage form was developed. Chromatographic separation was carried on Thermosil C18 (100*4.6 mm, 5), with a mobile phase composed of Methanol and pH 7 Triethylamine (70:30 V/V) at an absorption maxima 260 nm. Linearity for detector response was observed in the concentration range of 10-50 g/ml for Balofloxacin.. Correlation coefficient found to be 0.999. Retention time was found to be 3.706 min Balofloxacin.. Percent recovery studies were found in the range 99.0 – 101.0 % of test concentration. Drug product was exposed to acid, base, heat, oxidation and photolytic stress conditions and the samples were analysed by the proposed validated method. Results of the analysis were validated statistically and by recovery studies. The developed method was found to be precise for the determination of Balofloxacin in pure and its pharmaceutical dosage form.

Keywords: Balofloxacin, RP-HPLC, Stress degradation, Accuracy, Precision

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Spectrophotometric Estimation of Fexofenadine in Pharmaceutical Formulations by Using 3-Methyl Benzothiazolinone-2-Hydrazone

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Abstract- A simple, accurate, rapid and sensitive method has been developed for the estimation of fexofenadine in pharmaceutical dosage form. The method is processed based on oxidative coupling reaction of MBTH and Ferric chloride with fexofenadine to form greenish blue coloured chromogen showing maximum absorbance at 624nm. The method has been statistically examined and was proved to be high precise and accurate. The selective method is economical and sensitive for determination of fexofenadine in bulk drug and in formulation.

Keywords: Spectrophotometry, fexofenadine, MBTH (3-methyl benzothiozolinonehydrazone), Ferric chloride.

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A STUDY ON WORK LIFE BALANCE AMONG EMPLOYEES IN NCS SUGARS LIMITED, VIZIANAGARAM

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ABSTRACT

Work-life balance is concept creating and maintaining supportive and healthy work environments, which will facilitate employees to have balance between work and personal obligations and thus make stronger employee loyalty and productivity. Work life balance is gradually becoming a common talk. When employees go back to their homes, they should not carry any organizational stress with them. An individual has two roles to play personal and professional; each role having different set of demands. When such role demands overlap, multiple problems are faced leading to losses for all concerned: the individual, the family, the organization and the society. This investigative research is an attempt to study the work life balance issues with reference to employees in NCS sugars. The results derived from data analysis disclose significant results with respect to work life balance. The study has broad implications for industry in particular. The findings have significant implications for understanding, interpreting, and utilizing the contemporary work at sugar industries.

Key Words: work-life balance, multiple factors, work life equilibrium, employee loyalty

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INTRODUCTION

An increasing number of studies have projected the importance of work–life balance. This analyzed the existing concern within custom and organizations about the impact of multiple roles on the health and well-being of professional and its implications regarding work, family performance, and citizen's role in society. The following factors effected the experience of

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A STUDY ON NON PERFORMANCE ASSETS IMPACT ON FINANCIAL SUSTAINABILITY OF COMMERCIAL BANKS IN INDIA

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Abstract: *This study attempts to measures the financial sustainability towards NPA of the Commercial Banks in India. Analytical frame of the NPA emerged as an upsetting intimidation in the nation sending adverse signals on the sustainability and insurability of the debt burden banks. Among many desirable well-functioning characteristics of the financial system, management of NPA is a major one. The Government and RBI initiated multiple steps to control the upsurge in NPA, but did not result expectedly. As NPA can't be completely removed from the financial system, its furtherance should be resisted at the earliest to overcome excessive future financial erosion of capital. The study is based on secondary data collected from annual reports of department of Banking Supervision, RBI for the study period i.e. 2005 to 2017.To test the hypothesis the study has been worked on Student t-test by using SPSS. The paper has discussed on three group of banks; nationalized banks, state bank its India and its associates and public-sector banks further these banks are classified into three sectors; priority sector, non-priority sector and public-sector banks in India for scrutiny and revealed that the upward trend in NPA is persistently strong, eroding the profitability of the banks.*

Key words: *Operational Performance, Profitability, Non-Performing Assets, financial sustainability.*

INTRODUCTION: Assets which generate income are called performing assets and but those do not generate income are called non-performing

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Abstract: *This study attempts to measures the financial sustainability towards NPA of the Commercial Banks in India. Analytical frame of the NPA emerged as an upsetting intimidation in the nation sending adverse signals on the sustainability and insurability of the debt burden banks. Among many desirable well-functioning characteristics of the financial system, management of NPA is a major one. The Government and RBI initiated multiple steps to control the upsurge in NPA, but did not result expectedly. As NPA can't be completely removed from the financial system, its furtherance should be resisted at the earliest to overcome excessive future financial erosion of capital. The study is based on secondary data collected from annual reports of department of Banking Supervision, RBI for the study period i.e. 2005 to 2017.To test the hypothesis the study has been worked on Student t-test by using SPSS. The paper has discussed on three group of banks; nationalized banks, state bank its India and its associates and public-sector banks further these banks are classified into three sectors; priority sector, non-priority sector and public-sector banks in India for scrutiny and revealed that the upward trend in NPA is persistently strong, eroding the profitability of the banks.*

Key words: *Operational Performance, Profitability, Non-Performing Assets, financial sustainability.*

INTRODUCTION: Assets which generate income are called performing assets and but those do not generate income are called non-performing

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Employee Motivation and its Factors: An Analytical Study Conducted in an Educational Institution

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Abstract- Motivated employees play a vital role in transforming an organization from good to great. They are productive and also infuse life into the organization. Their energy and commitment is infectious, which drives other employees also to deliver more. Employee motivation is a factor/s that induces an employee to own tasks and responsibilities and deliver results with commitment and ownership. Motivating employees in educational institutions is very crucial as employees bear the responsibility of imparting knowledge, skills, values and ethics to the students who will be the flag bearers of the future generation. This study is an attempt to assess employee motivation in one of the of the fastest emerging private engineering colleges in Visakhapatnam and tried to identify the factors that have a bearing on employee motivation. This study makes use of self-completion questionnaires with 50 employees. The study results may be useful for the all the stakeholders of an educational institution for better understanding of motivational factors, which are essential to keep the employees motivated.

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RESEARCH ARTICLE OPEN ACCESS

An Experimental Study on Mechanical Properties of Hooked Steel Fiberous Concrete

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ABSTRACT: Concrete being brittle is weak in tension. The addition of Steel Fibers in concrete have greatly progresses its Compressive Strength, Split tensile strength as well as Flexural Strength . The use of dissimilar types of Fibers & their orientation in the cement Concrete matrix have shown positive responses between the scholars. Fiber is easy available material. The crack failures can be reduced by means of the steel fibers in concrete. These fibers are thin, short and distributed randomly all over the concrete member of different aspect ratios (L/d ratio). Out of all different types of Fibers, steel Fibers are widely used because steel has high elasticity Modulus , high elongation, high tensile strength and the bond between steel and the Fiber is enormous. In This present experimental investigation was carried out to investigate the influence of steel Fibers on physical and mechanical properties of concrete, covering cold strained carbon steel Fibers of hooked end type having aspect ratio of 50 with diameter of 0.6mm and length 30mm with following percentages of 0.5%, 1%, 1.5% and 2.0% fraction is added to the concrete. Concrete is evaluated for compressive, split tensile and flexural strength at the ages of 7, 14 and 28 days, with the adding of 1.5% of Steel Fibers, test results show the maximum compressive Strength, split tensile Strength and flexural strength, it becomes the optimal value. Moreover, the results confirmed that the fibrous steel reinforced concrete reduce cracking and improves flexure.

Keywords: Compressive strength, Hooked steel fiber, aggregate and cement

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I. INTRODUCTION

In present scenario concrete is one of the most widely used materials in the construction field. Usually, it is associated with ordinary Portland cement which is the main Component for making concrete[1]. Cement offers excellent performance as a bind-ing material in concrete. So the demand for cement rises day by day due to development of infrastructure all over the world. On another hand, cement manufacturing Companies are increases from past decades to till now

The concrete without any Fibers will develop the cracks due to plastic shrink-age, drying shrinkage and other reasons of changes in volume of concrete[2]. The development of these micro cracks causes elastic deformation of concrete. Plain concrete is a brittle material and having the values of modulus of rupture and strain capacity is low. In order to meet the required values of flexural strength and enhances the strain capacity of the plain concrete, the Fibers are being used in normal concrete[3]. The adding of Fibers in the plain concrete will control the cracking due shrinkage and also reduce the bleeding of water.

Plain concrete processes a very low tensile strength, limited ductility, and little resistance to cracking. Internal micro cracks are inherently present in the concrete and its poor tensile strength is due to the propagation of such micro cracks, eventually leading to brittle failure of the concrete.The most widely accepted remedy to this flexural weakness of concrete is the conventional reinforcement with high strength steel. Restraining techniques are also used to. Although these methods provide tensile strength to members, they however do not increase the inherent tensile strength of concrete itself.

Fiber Reinforced Concrete

Fiber is a small piece of reinforcing material possessing certain characteristics properties. They can be circular or flat. The fiber is often described by a convenient parameter called "aspect ratio". The aspect ratio of the fiber is the ratio of its length to its diameter. Typical aspect

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Chapter 48
Inventory and Monitoring of Glacial Lakes in Himalayan Region Using Geo Spatial Technologies

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P. Satyanarayana, E. Siva Shankar, E. Amminedu and N. Victor Babu

Abstract The glaciers consist of a huge amount of perpetual snow and ice and are found to create many glacial lakes. Abrupt release of large amounts of stored water in these lakes results in a catastrophic outburst flood. Several such incidents of flash floods were reported all over the world and also in Himalayas region of India causing huge economic losses. To mitigate the impact of these types of events, prior knowledge about the location, the areal extent and the volume of these lakes is very essential. The location of these glacial lakes in rugged and remote terrain makes it difficult to monitor them manually. Remote sensing plays a vital role in creating inventories and monitoring of the glacial lakes quickly and accurately due to wider coverage and repetivity. The satellite images provide greater details for the evaluation of physical conditions of the area. This paper discusses a case study on the inventory and monitoring of glacial lakes or water bodies in Sutlej basin using satellite remote sensing. The study area is Sutlej basin from its origin to Bhakra dam situated in Western Himalayas. This basin is highly rugged terrain with abundant natural water resource in the form of snow pack and glacial lakes. The Inventory of glacial lakes or water bodies was done using Indian Remote Sensing LISS-III data and monitoring using AWiFS data. A total of 197 lakes or water bodies have been identified whose water spread area is greater than 2 ha. 40 lakes with area greater than 10 ha were monitored during June–August 2007. The change in water spread area statistics was observed in 23 lakes with a variation of greater than ±10%.

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Chapter 67
A Geospatial Study on Visakhapatnam Smart City Based on Hydrogeochemistry Analysis

Neela Victor Babu, Peddada Jagadeeswara Rao, Suribabu Boyidi and Sridhar Bendalam

Abstract Groundwater tainting because of ocean water interruption is a surely understood issue in Madhurawada panchayat (group of villages). This investigation endeavor to give answers the hydrogeochemistry idea of groundwater synthetic parameters in Madhurawada Panchayat. In the ponder we apply existing field bore well and burrowed well perception which join with research facility water test examination as indicated by WHO and BIS norms and obviously aberrant addressing with the villager. The outcome has demonstrated the sullying level of EC, TDS, TH, Ca^+, Na, Cl^-, F^-, and Fe in the Paradesipalem, Boravanipalem, KommadiJn, Marikavalasa, Thotlakonda, Jodugullapalem and Boddapalem in both bore well and burrowed wells over the farthest point of reasonable farthest point for drinking limit. Calcium and magnesium are the predominant cations display in groundwater beside sodium and chloride in this district. In this manner, order ought to take at the season of groundwater utilization.

Keywords Ground water samples · GIS · GVMC · Hydrogeochemistry

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Earth Quake Resistant Building Design by Consider Bracings and Shear Wall System in E-tabs Software

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ABSTRACT:

An earthquake (also referred to as a quake, tremor or temblor) is the shaking of the surface of the Earth, because of the surprising release of energy inside the Earth's lithosphere that creates seismic waves. Earthquakes can range in length from the ones that are so weak that they can't be felt to those violent enough to toss humans around and break whole cities. The seismicity, or seismic hobby, of a place is the frequency, type and length of earthquakes skilled over a time frame. The word tremor is also used for non-earthquake seismic rumbling. Earthquake-resistant structures are systems designed to protect buildings from earthquakes. While no shape may be absolutely resistant to damage from earthquakes, the goal of earthquake-resistant construction is to erect systems that fare higher at some stage in seismic pastime than their conventional counterparts. According to constructing codes, earthquake-resistant systems are supposed to resist the largest earthquake of a certain probability this is likely to arise at their area. This way the lack of lifestyles must be minimized via preventing crumble of the homes for uncommon earthquakes even as the loss of the functionality need to be restricted for greater common ones. Now a day's steel bracings technique and shear wall systems are generally using for designing of earth quake resistant structure due to simple construction methods, easy to install and they are reduces the deflection and shear in past studies the earth quack resistant structure is designed by using steel bracings or shear wall systems in present study a comparison made between these two systems along with general building in high seismic zone. In the present study a G+10 story is modeled by using E-TABS software and analyzed in push over analysis and the comparison is made between the general building, steel building and shear wall buildings to design the earth quake resistant structures design. The results like story drift, story shear, story moment, building torsion, time period, and model stiffness were compared.

KEY WORDS: E-TABS, seismic analysis, shear wall, pushover, Torsion, Stiffness.

INTRODUCTION:

Earthquakes are certainly one of nature's most outstanding risks to lifestyles in the world and feature decimated incalculable city regions and cities on for all intents and functions each landmass. They are one in all man's most dreaded normal marvels due to actual seismic tremors turning in highly Immediate pulverization of systems and specific structures. Furthermore, the harm caused by Earthquakes is at the whole linked with synthetic systems. As inside the instances of avalanches, seismic tremors likewise purpose passing with the aid of the harm they instigate in structures, as an example, structures, dams, spans and special works of man. Sadly massive numbers of Earthquakes deliver almost no or no observe before taking place and this is one purpose why Earthquake building is complicated. Nowadays the townhouse building is basic work of the social advance of the province. Everyday new of living arrangements financially, rapidly and satisfying the prerequisites of the gathering specialists and creators do the crease

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RESEARCH ARTICLE

Condensation Methods for the Determination of Methoxamine Hydro Chloride in Pure and Pharmaceutical Formulations

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ABSTRACT:

Two visible spectrophotometric methods were developed Aand B for the determination of Darunavir in pure and pharmaceutical formulations. The methods are based on condensation reaction with PDAB (Method-A) and ONB (Method-B) in presence of acidic medium with the primary amine group in DNV. The coloured products exhibit absorption λ_{max} at 639 nm and 452nm for methods A and B respectively. Regression analysis of Beer-Lambert plots showed good correlation in the concentration ranges 10-60μg/ml, 50-300μg/ml, correlation co-efficients are 0.9983, 0.9989; Sandell's sensitivities are 9.9833×10^{-3}, 3.0456×10^{-2} (1 mole cm^{-1}); and molar absorptivity values are 5.4857×10^{4}, 1.7981×10^{4} (μg cm^{-2}) for methods-Aand B respectively. The proposed methods are applied to commercial available formulations and the results are statistically compared with those obtained by the UV reference method and validated by recovery studies. The results are found satisfactory and reproducible. These methods are applied successfully for the estimation of the DNV in the presence of other ingredients that are usually present in formulations. These methods offer the advantages of rapidity, simplicity and sensitivity and low cost without the need for expensive instrumentation and reagents.

KEYWORDS: Condensation, PDAB, ONB, Regression Analysis.

INTRODUCTION:

Methoxamine hydrochloride (MHC), chemically 2-amino-1-(2,5-dimethoxyphenyl) propan-1-ol (Fig.1), is relatively selective α_1-adrenoreceptor agonist producing an increasing in peripheral vascular resistance1,2. MHC is a potent sympathomimetic that increases both systolic and diastolic blood pressure. It is used in the treatment of some hypotensive patients, particularly those with paroxysmal a trial tachycardia (arrhythmia) andacts almost exclusively on alpha-adrenergic receptors.

Literature survey reveals few analytical methods were reported for the determination of MHC in biological fluids by HPLC with fluorescence detection3,4. Spectrophotometric method analysis 5,6. High performance liquid chromatography using pre-column derivatization of MHC with other cardiovascular drugs7. The analytical useful functional groups in MHC have not been fully exploited for designing suitable visible spectrophotometric methods and so still offer a scope to develop more visible spectrophotometric methods with better sensitivity, precision and accuracy.

No analytical methods that have been reported so far for the estimation of MHC by HPLC method. Hence, the author has made an attempt to develop a new, simple, economical, selective, accurate and precise stability indicating reverse phase high-performance liquid chromatographic method 8-12 with good sensitivity for assay of MHC in pharmaceutical dosage form. Indeed the established method was validated with respect to

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Review Article

Rudiments of Syncretism in Nayantara Sahgal's *Mistaken Identity*

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Abstract: In a country like India with diversified culture and social backdrop, syncretism with its delicate manure happens to be a significant phenomenon that has been given due emphasis by Nayantara Sahgal in her novel, *Mistaken Identity*. Sahgal has not only shown utmost concern for a united nation but has tried to bring to the foreground its crude limitations in a multicultural society. Sahgal's prime focus has been for a unification of the humanistic values with the divergent features of various religions prevailed in this subcontinent through the character, Bhushan, the protagonist in the novel. This paper has tried to make a resilient attempt in exploring the unique racial and ethnic prospect replicating the exceptional irreligious and syncretic mosaic that yearns to prevail amidst the multicultural society of human habitation.

Keywords: Ethnicity, syncretic aspects, fundamentalism, egalitarian culture, cognizance of transformation.

INTRODUCTION

The novel *Mistaken Identity* is set up during the time of the twilight years of the British rule in India. Sahgal desires towards explicating a unique platform thereby witnessing the several acrimonious multiethnic rudiments merging into manufacturing a distinctive, harmonious and syncretic assortment that brings under one umbrella the diverse groups united. In this novel, Bhushan, the protagonist functions as the representative of the novelist in this aspect. Bhushan implores for the recognition of the synthesis in the numerous ethnic aspects which eventually paves its way towards an amalgamated culture whereby the different diverged entities find themselves intimate.

In the novel, *Mistaken Identity*, Nayantara Sahgal is very optimistic with regard to envisaging the syncretic aspects of the multiethnic apparition that embraces the inclusive culture of India in an amicable way.

A fine definition of Syncretism can be put forth as the amalgamation of unlike forms of belief or practice. In the words of Burman [1],

> The rise of fundamentalism in recent years has obliterated the deep rooted syncretism in Indian culture. In India, few have studied the syncretic phenomenon of local religion though many have studied it in terms of formation of composite culture.

Sahgal views that there is every possibility of an inclusive unity amid the cohorts of diverse religions that would certainly concrete the way for a magnificent India. Sahgal envisions an India as a country in which her associated nationals are empathetic and harmonious with an egalitarian, secular and expanded cultural outfit. A unique glimpse of creative susceptibility in nurturing the spirit of national consciousness is apparently vigilant in Sahgal; moreover she also attempts to mirror the cognizance of transformation and the extraordinary, impulsive means in which the character of the individuals have been shaped thereby having a correlational effect with the rage of the collective life as well as the contending edifices of human ethics and destination.

In the novel, *Mistaken Identity*, an exceptional merger of the distinctive chronological and ethnic ambivalences of the Indian convention has been brought to the foreground. India with its fundamental feature of secular democratic concern as well as the reservoir of rich, age old cultural heritage which is many millennia young appeals for the harmonious co-existence of different religions under the remarkable umbrella of human passion and compassion. India has been pictured in its twilight years of imperialistic rule in

Three Phase Multilevel Inverter Using Switched Capacitor Units Fed Induction Motor Drive

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Abstract- Multilevel inverters have been attracting in favor of academia as well as industry in the recent decade for high-power and medium-voltage energy control. In addition, they can synthesize switched waveforms with lower levels of harmonic distortion than an equivalently rated two-level converter. The multilevel concept is used to decrease the harmonic distortion in the output waveform without decreasing the inverter power output. The proposed inverter can output more numbers of voltage levels in the same number of switching devices by using this conversion. The number of gate driving circuits is reduced, which leads to the reduction of the size and power consumption in the driving circuits. The total harmonic of the output waveform is also reduced. The proposed inverter outputs larger voltage than the input voltage by switching the capacitors in series and in parallel. The maximum output voltage is determined by the number of the capacitors. In this paper two new topologies have been proposed for multilevel inverters. The proposed topologies consist of a combination of the conventional series and the switched capacitor inverter units. The proposed method introduces 17, 25 levels and three phase 25level Inverter fed Induction Motor drive. With the use of high level inverter, resolution is increase and also the harmonics is highly reduced. The simulation results are presented by using Matlab/simulink software.

Keywords: Multilevel inverter, series inverters, series- parallel connection, switched capacitor, induction motor.

I. Introduction

Power electronic converters, particularly dc/air conditioning PWM inverters have been broadening their scope of utilization in industry since they give decreased vitality utilization, better framework effectiveness, enhanced nature of item, great upkeep, etc. For a medium voltage matrix, it is troublesome to associate just a single power semiconductor switches straightforwardly [1]. Therefore, a staggered power converter structure has been presented as an option in high power and medium voltage circumstances, for example, laminators, factories, transports, siphons, fans, blowers, blowers, etc. As a practical arrangement, staggered converter accomplishes high power appraisals, as well as empowers the utilization of low power application in sustainable power sources, for example, photovoltaic, wind, and energy units which can be effectively interfaced to a staggered converter framework for a powerful application. The most widely recognized beginning use of staggered converters has been in footing, both in trains and track-side static converters [2].
Recently, some new topologies of staggered inverters have risen. This incorporates summed up staggered inverters, blended staggered Introduction 4 inverters [10], cross breed staggered inverters [11] and delicate exchanged staggered inverters. These staggered inverters can expand appraised inverter voltage and power by expanding the quantity of voltage levels. They can likewise build proportional exchanging recurrence without the expansion of real

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Optimization & Analysis Techniques of switched-capacitor DC-DC Converters

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Abstract- A switched-capacitor (SC) dc–dc converter's steady-state performance through evaluation of its output impedance. This analysis method has been verified through simulation and experimentation. The simple evaluation developed permits evaluation of the capacitor sizes to meet a parameters such as a total capacitance or total energy storage limit, and also permits optimization of the switch sizes subject to constraints on total switch conductance or total switch volt-ampere (V-A) products. These techniques allow comparison among several switched-capacitor topologies, and SC converters with conventional magnetic- based dc–dc converter circuits, in the context of various application settings. Significantly, the performance (based on conduction loss) of a ladder-type converter is found to be superior to that of a conventional magnetic-based converter for medium to high conversion ratios.

Keywords: Analysis, dc-dc converter, output impedance, switched capacitor

1. INTRODUCTION

This thesis is used to evaluate steady state performance of a switched capacitor dc- dc converter by determining its output impedance. This thesis builds up a circuit, which can be utilized to access both the converter proficiency and produce output as a component of load for an expansive class of SC converters.

This method created grants improvement of the capacitor sizes to meet a limitation. These improvements are done among several switched capacitor topologies. These optimizations at that point allow comparison among a few switched capacitor topologies, and correlations of SC converters with traditional magnetic based dc– dc converter circuits. The execution of a ladder type converter is observed to be better than that of a traditional boost converter for medium to high change proportions.

2. Switched Capacitor Converter Impedance Analysis

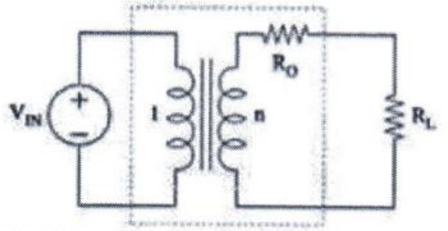

Figure 1. Model of an idealized switched capacitor converter

Optimization & Analysis Techniques of switched-capacitor DC-DC Converters

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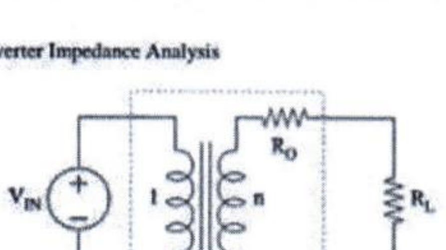

Figure 1. Model of an idealized switched capacitor converter

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Bidirectional Single-Stage Grid-Connected Inverter for a Battery Energy Storage System

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Abstract—The main objective of this paper is for the battery energy storage system to propose a bidirectional single-stage grid-connected inverter (BSG inverter). This is composed of multiple bidirectional buck-boost type dc-dc converters (BBCs) and a dc-ac unfolder. single-stage power conversion, low battery and dc-bus voltages, pulsating charging/discharging currents, and individual power control for each battery module are the advantages of the proposed BSG-inverter include:. Therefore, the equalization, lifetime extension, and capacity flexibility of the battery energy storage system can be achieved. Based on the developed equations, the power flow of the battery system can be controlled without the need of input current sensor. Also, with the interleaved operation between BBCs, the current ripple of the output inductor can be reduced too. The computer simulations and hardware experimental results are shown to verify the performance of the proposed BSG-inverter.

I. INTRODUCTION

BECAUSE of the fossil fuel exhaustion and global warming issue, renewable energies such as the photovoltaic (PV) power and wind turbines are more and more popular recently. However, the fluctuations of the high penetration renewable energy will cause the negative impact to the grid voltage and frequency stabilization. A battery energy storage system is a promising candidate to increase the penetration rate of the renewable energy. For the microgrid application, the battery energy storage system is essential not only for controlling and managing the energy of distributed generation units such as PVs, wind turbines, and microturbines for the stability of the power system, but also for protecting loads from grid fault conditions. As shown in Fig. 1, the conventional battery energy storage system consists of a battery array, which is formed by many battery modules connected in series or parallel, and a bidirectional grid-tied dc–ac inverter as a full-bridge inverter [1]–[3

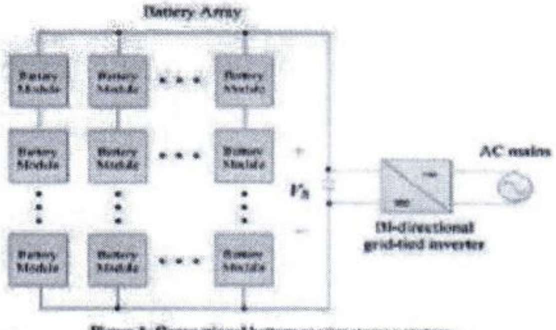

Figure 1. Conventional battery energy storage system

Circuit simplicity is the main advantage of this type of battery energy storage system but the total power capacity may be easily reduced by a particular overcharging/discharging battery module due to the battery tolerance, unequal battery losses, and so on. In order to maximize energy storage, the voltage of the individual battery module connected in series to form a dc bus as the input of the grid-tied inverter must be equalized with each other. The general solution to solve the battery capacity reduction problem is to use extra balancing circuit to connect each battery module and balance the charge of all battery modules. However, the balancing circuit may result in the reduction of total efficiency and the increase of cost and circuit complexity.

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Fe_2O_3/RGO nanocomposite photocatalyst: Effective degradation of 4-Nitrophenol

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ABSTRACT

We synthesized Fe_2O_3/RGO nanocomposite (NC) via simple and green hydrothermal method. The structure, morphology and thermal stability of prepared NC was studied using XRD, FESEM, EDS, TGA, FTIR and UV-DRS. As-synthesized NC was tested for photocatalytic activity by the degradation of model organic pollutant, 4-Nitrophenol (4-NP) under visible light irradiation. The effect of pH and H_2O_2 were studied on photocatalytic degradation of 4-NP by Fe_2O_3/RGO NC. This NC as benign photocatalyst for completely degraded 4-NP in 50 min. We stated that H_2O_2 played a key role in photocatalytic degradation of dye pollutant under visible light irradiation with the help of composite which was prepared by hydrothermal route. However H_2O_2 acts as promising oxidant due to its high efficiency, low cost and produce harmless co-additives. Moreover H_2O_2 can serve as electron scavengers and generates the hydroxyl radicals.

1. Introduction

Due to rapid industrialization and urbanisation, modern technology causes water contamination. Especially, industries release various pollutants like organic dyes, pesticides and fungicides with effluents that are toxic, carcinogenic and hazardous to health. Hence, this waste water must be processed before disposal. Nowadays, numerous techniques currently in use for wastewater treatment most are inefficient, cost- or/and energy-intensive and result in the generation of large amounts of sludge [1, 2]. In the past two decades, an advanced oxidation process (AOP) has been developed in water treatment. AOP is a photocatalytic process that degrades toxic chemicals into CO_2, H_2O and non-toxic compounds under light in water. Metal oxides and their composites as heterogeneous photocatalysts have been reported recently [3–5].

F[illegible]s n-type semiconducting with an indirect band gap of 2.2 eV and it has attractive wide applications in pigment, gas sensors and catalysis [6-8]. However, the bandgap of Fe_2O_3 was reduced for better photocatalytic activity by formation of composite with tremendous properties materials like Graphene or polymer. Graphene is a sp^2 hybridised and zero bandgap with excellent thermal, structural and catalytic properties, thus it is a promising material for wider applications in gas sensors, catalysis, energy storage and imaging [9-10]. To best of my knowledge, very few reports on preparation of Fe_2O_3/RGO for photocatalytic activity and this composite was synthesized by various methods such as co-precipitation, microwave, hydrothermal, sol-gel and solvothermal [11]. Among all, hydrothermal is green approach and simple for preparation of nanomaterials. Hence, we report a synthesis of magnetic reduced graphene oxide (Fe_2O_3/RGO) nanocomposite via hydrothermal route for photocatalytic degradation of 4-Nitrophenol under visible light irradiation.

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2. Experimental

2.1 Materials

Graphite flakes, potassium permanganate ($KMnO_4$), Ferric chloride ($FeCl_3$ $6H_2O$), phosphoric acid (H_3PO_4), sulphuric acid (H_2SO_4), hydrogen peroxide (H_2O_2) and hydrochloric acid (HCl) were procured from Sigma Aldrich, India. 4-Nitrophenol was received from Himedia, India and Milli Q water was used throughout all the experiments.

2.2 Synthesis of GO

Graphene oxide (GO) was prepared by modified Hummers method [12]. In briefly, 0.25 g of graphite flakes, 27 mL of H_2SO_4 and 3 mL of H_3PO_4 were mixed and stirred for several minutes. 1.32 g of $KMnO_4$ was then added slowly into the solution. This mixture was stirred for 6 hours until the solution became dark green. To eliminate excess of $KMnO_4$, 0.7 mL of H_2O_2 was dropped slowly and stirred for 10 minutes. The exothermic reaction occurred and let it to cool down. 10 ml of HCl and 30mL of Milli Q water was added and centrifuged. Then, the supernatant was decanted and then rewashed again with HCl and Milli Q for 3 times. Finally the obtained GO was dried in oven at The washed GO solution was dried using oven at 90° C for 24 h.

Synthesis of Fe_2O_3/RGO composite

In typical synthesis, 10 mL of $FeCl_3$ $6H_2O$ (0.1 M) solution and 10 mL NaOH aqueous solution were taken into 100 ml beaker, then added to 50 mg GO under vigorous stirring for 30 min, and the slurry solutions were poured into Teflon lined autoclaves and hydrothermally heated at 180°C for 24 h. The yield was washed with water then followed by ethanol, centrifuged and dried at 60° C in oven at overnight. The same procedure was used for the synthesis of pure Fe_2O_3 without adding of GO.

2.3 Characterization of prepared samples

The prepared catalysts were characterized using Powder X-ray diffraction (XRD) (D8 Focus, Bruker instrument, Germany) with Cu Kα radiation (λ=1.5406 A°), 2θ ranges from 10° to 90° with scanning speed of 0.02° S^{-1} to identify the crystalline phase of samples. Fourier Transform Infrared (FTIR)

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ZnO/RGO nanocomposite via hydrothermal route for photocatalytic degradation of dyes in presence of visible light

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Abstract

The organic dye pollutants such as Congo red (CR) and Eosin yellow (EY) contaminate the water. Hence there is need to develop composite for removal of dyes. In this manuscript, we reported the reduced graphene oxide based zinc oxide (ZnO/RGO) nanocomposite (NC) was synthesized by simple and green hydrothermal method. As-prepared nanocomposite was characterized by XRD, FESEM, EDS, TGA, FTIR and UV-DRS to study its structure, morphology and thermal stability. As synthesized composite was executed excellent photocatalytic activity by the degradation of organic dye pollutants, CR and EY under visible light irradiation. The effect of pH, concentration dye solution and catalyst dose were studied to optimise. The optimised conditions for degradation of CR using ZnO/RGO composite are 0.05g catalyst dose, pH-3 and initial concentration 10 ppm of dye solution. From these results, this composite served as benign photocatalyst for completely degraded CR in 120 min and EY in 100 min respectively.

Keywords: ZnO/RGO nanocomposite, hydrothermal, photocatalytic, Congo red (CR) and Eosin yellow (EY)

1. Introduction

Due to rapid industrialization and urbanisation, modern technology causes water contamination. Especially, industries release various pollutants like organic dyes, pesticides and fungicides with effluents that are toxic, carcinogenic and hazardous to health. Therefore, development of a simple method for determination with high selectivity and sensitivity is very important in many fields [1, 2]. All kinds of methods have been developed. The advanced oxidation processes (AOPs) has most advantages such as simple, fast, low-cost, and portable which degrades dye solution into non-toxic compounds like CO_2, H_2O in the presence of light. Adsorptions of organic pollutants are a necessary condition for the oxidation of these substances to take place on the surface of photocatalysts [3]. Semiconductors like TiO_2, CuO, Fe_2O_3, SnO_2 and ZnO emerged as benign photocatalysts in AOPs. Recently metal oxides and their composites as heterogeneous photocatalysts have been reported [4, 6].

Zinc oxide (ZnO) is a semiconductor with wide band gap of 3.37 eV and has been used considerably for its catalytic, electrical, optoelectronic, and photochemical properties. ZnO nanostructures have a great attention in catalytic reaction process due to their large surface area and high catalytic activity [7]. But, it has drawback in photocatalysis due to recombination of electron-hole pair is faster. In order to overcome this, additional material is essential to enhance this gap. Graphene oxide (GO) has great attention to researchers because of its properties and it slower the recombination of electron-hole pair [8].

In this paper, we report the hydrothermally preparation of ZnO/RGO nanocomposite for photocatalytic degradation of dyes like Congo red (CR) and Eosin yellow (EY) under visible light irradiation.

2. Experimental section

2.1 Materials

Graphite flakes, sodium nitrate ($NaNO_3$), potassium permanganate ($KMnO_4$), hydrogen peroxide (H_2O_2), sulphuric acid (H_2SO_4), hydrochloric acid (HCl), zinc acetate